

AmradAI – Intelligent Skin Disease Detection

AI-Based image analysis for early skin disease detection

Student: [Your Name] · Course: [Course Name] · University:
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Introduction: Why this matters

Skin diseases are common worldwide. Early detection drastically improves outcomes – especially for conditions like melanoma and other skin cancers. AmradAI uses AI to analyze user-submitted skin photos and surface potential risks, making early screening more accessible.

Problem Statement

Missed early signs

Many people overlook subtle changes in moles or lesions until they progress.

Limited access

Dermatology care can be expensive or unavailable in many regions.

Hidden symptoms

Early-stage skin cancer often appears innocuous and is easily missed by non-specialists.



Our Solution: AmradAI

AmradAI is an AI-powered website where users upload a photo of a skin spot or mole. The platform analyzes the image using trained models and returns an easy-to-understand assessment:



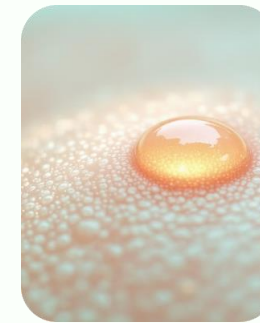
Normal mole

Low concern – routine monitoring recommended.



Suspicious condition

Possible abnormality – consider dermatologist consult.



Possible early skin cancer

High concern – urgent medical evaluation advised.

System Workflow — Technical Overview

Preprocessing

Normalize, crop, and augment images for model input.

CNN Classification

Model outputs Normal, Suspicious, or Possible cancer.

Upload Image

User submits a skin image via the web app.

User Feedback

Provide results, confidence, and recommended next steps.

Core technologies:

- Artificial Intelligence & Convolutional Neural Networks
- Machine Learning for classification and confidence scoring
- Web application stack for uploading, processing, and presenting results
- Data analysis for model validation and monitoring



Model & Data Approach

We train CNN models on curated, anonymized datasets with diverse skin tones and lesion types. Key steps: data cleaning, augmentation, class-balance strategies, cross-validation, and performance metrics (accuracy, precision, recall, AUC).



Privacy-first

Images anonymized; users control uploads and deletion.



Bias mitigation

Balanced dataset sampling and fairness evaluation across skin tones.



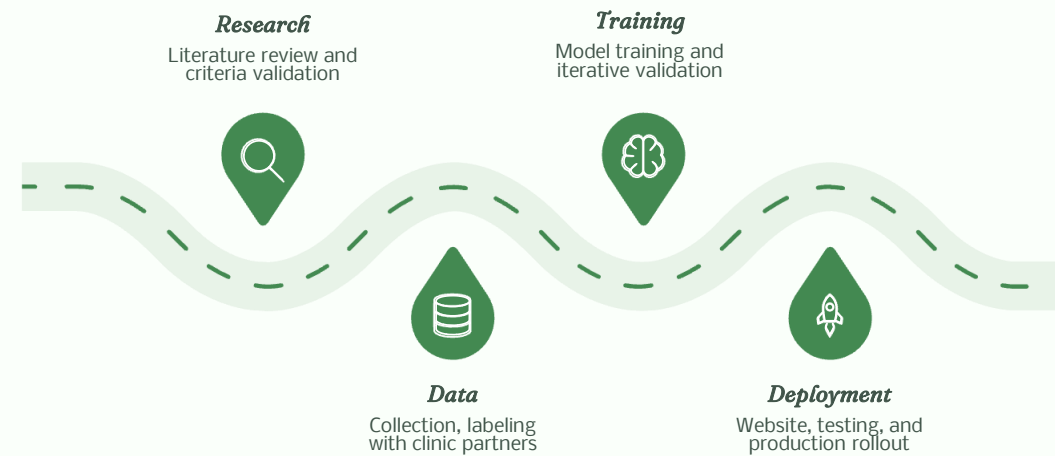
Continuous learning

Periodic retraining with new validated cases to improve accuracy.

Project Roadmap



Development Timeline



Milestones:

- Research & literature review – validate clinical criteria
- Data collection & labeling – partner with clinics, ensure diversity
- Model training & validation – iterate on architectures and metrics
- Frontend & backend development – secure, responsive web app
- Testing – clinical validation, user testing, and regulatory review
- Deployment & monitoring – CI/CD, model drift detection

Skills Acquired



Artificial Intelligence

Designing and training classification models; evaluating model metrics.



Web Development

Implementing secure upload flows, APIs, and responsive UI components.



Data Processing

Cleaning, augmentation, and maintaining datasets for fairness and performance.



UI/UX Design

Creating accessible, user-friendly interfaces for non-expert users.

Conclusion & Impact

AmradAI demonstrates how AI can support earlier detection of skin diseases and encourage timely medical care. It is not a replacement for clinicians – it's a screening and triage tool that increases access, raises awareness, and assists clinicians with preliminary analysis.

📌 Note: Clinical validation, regulatory compliance, and ethical safeguards are essential before any public deployment.



Thank You

Questions?

Contact: [Your Email] · Project repo: [Repository Link]



Next steps

Clinical partnership, extended dataset collection, formal evaluation study.



Call to action

Seeking collaborators: dermatology partners, data scientists, and UX researchers.